

TYNEMOUTH, United Kingdom

WMO: 032620

Lat: 55.017N

Lon: 1.417W

Elev: 30

StdP: 100.97

Time Zone: 0.00 (EUW)

Period: 86-99

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.5	-0.2	-4.5	2.6	0.4	-3.4	2.9	1.5	23.1	5.6	21.4	4.6	6.6	280	0.805

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.7	21.0	16.3	19.6	15.8	18.6	15.1	17.6	19.7	16.7	18.7	15.9	17.8	6.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.7	11.9	18.5	15.8	11.3	17.7	14.9	10.6	17.0	49.5	19.8	46.8	18.9	44.4	17.9	22.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.7	17.7	14.9	-4.2	25.0	2.1	2.9	-5.7	27.1	-6.9	28.8	-8.1	30.4	-9.6	32.5		
(5)			-4.6	19.3	2.1	1.5	-6.1	20.4	-7.3	21.3	-8.5	22.1	-10.0	23.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.4	5.0	5.0	6.4	7.1	10.1	12.3	15.0	14.9	12.8	10.3	7.3	5.8
	DBStd	4.38	2.71	3.03	2.72	2.64	2.42	2.32	1.92	2.29	2.23	2.64	2.59	2.81
	HDD10.0	795	157	142	117	95	29	5	0	0	3	29	86	132
	HDD18.3	3283	414	373	371	337	256	182	106	108	168	250	331	387
	CDD10.0	559	1	2	4	8	32	73	154	153	86	37	5	3
	CDD18.3	5	0	0	0	0	0	0	2	3	1	0	0	0
	CDH23.3	8	0	0	0	0	0	0	2	5	1	0	0	0
	CDH26.7	1	0	0	0	0	0	0	0	1	0	0	0	0
Wind	WSAvg	7.0	8.5	8.4	7.9	7.4	6.2	6.0	5.6	5.9	6.5	7.1	7.4	7.7
Precipitation	PrecAvg	732	61	50	43	53	50	62	64	69	58	71	82	69
	PrecMax	1028	166	114	101	148	100	164	156	157	147	159	176	150
	PrecMin	542	18	8	10	8	14	15	10	13	17	19	29	31
	PrecStd	117	32	27	24	40	24	38	32	31	33	29	35	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	13.6	14.7	16.3	18.3	20.2	22.7	23.1	21.7	17.9	14.3	14.4
		MCWB	10.4	11.2	11.1	12.6	13.4	15.5	17.7	17.4	17.6	14.8	12.0	12.5
	2%	DB	11.6	12.3	12.9	14.3	16.5	18.4	20.9	20.9	18.7	16.6	13.2	12.4
		MCWB	9.5	10.0	9.8	11.2	12.5	14.5	16.3	16.5	15.2	14.0	11.2	10.7
	5%	DB	10.3	11.2	11.7	12.8	14.9	17.1	19.5	19.7	17.3	15.2	12.1	11.2
		MCWB	8.5	9.1	9.3	10.3	11.7	13.9	15.7	16.0	14.0	13.0	10.4	9.5
	10%	DB	8.9	9.8	10.5	11.3	13.7	15.9	18.3	18.6	16.2	14.0	11.0	9.9
		MCWB	7.4	8.0	8.3	9.1	11.2	13.2	14.8	15.5	13.4	11.9	9.6	8.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.7	11.6	11.5	13.5	14.5	16.7	18.6	18.9	18.2	15.5	12.4	12.7
		MCDB	12.5	13.3	14.0	15.8	17.1	19.4	22.0	21.6	21.1	17.7	13.9	14.4
	2%	WB	9.7	10.2	10.2	11.6	13.1	15.3	17.1	17.8	16.1	14.4	11.5	10.8
		MCDB	11.4	12.0	12.3	13.6	15.4	17.1	19.9	19.7	18.0	16.2	12.8	12.2
	5%	WB	8.6	9.3	9.5	10.5	12.2	14.5	16.3	17.0	14.8	13.3	10.7	9.6
		MCDB	10.1	10.9	11.4	12.5	14.2	16.4	18.5	18.6	16.4	14.8	11.8	11.0
	10%	WB	7.6	8.1	8.6	9.4	11.5	13.6	15.6	16.2	14.0	12.2	9.8	8.5
		MCDB	8.9	9.7	10.4	11.1	13.4	15.6	17.5	18.0	15.4	13.3	10.8	9.8
Mean Daily Temperature Range	5% DB	MDBR	3.6	3.8	4.5	4.1	4.1	3.9	4.5	4.7	4.7	4.2	3.7	3.6
		MCDBR	5.3	4.8	6.0	6.7	6.6	6.6	6.9	6.7	6.7	5.8	4.8	4.8
		MCWBR	5.1	4.1	4.5	4.8	4.0	3.6	4.1	3.7	4.5	4.4	4.4	4.5
	5% WB	MCDBR	5.3	4.9	5.8	6.5	6.0	5.6	5.9	5.3	6.1	5.3	4.4	4.8
		MCWBR	5.1	4.3	4.7	4.9	4.0	3.8	4.0	3.6	4.8	4.4	4.2	4.6
Clear-Sky Solar Irradiance	taub	0.318	0.335	0.366	0.390	0.387	0.390	0.391	0.382	0.362	0.345	0.319	0.302	
	taud	2.321	2.388	2.319	2.277	2.340	2.359	2.362	2.401	2.458	2.458	2.377	2.287	
	Ebn at Noon	610	745	800	831	856	857	847	833	803	726	608	537	
	Edh at Noon	55	73	98	116	115	115	112	102	85	68	52	46	
All-Sky Solar Radiation	RadAvg	0.57	1.20	2.35	3.82	4.95	4.96	4.75	3.95	2.81	1.53	0.72	0.42	
	RadStd	0.07	0.13	0.29	0.36	0.26	0.40	0.37	0.26	0.24	0.12	0.06	0.04	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page